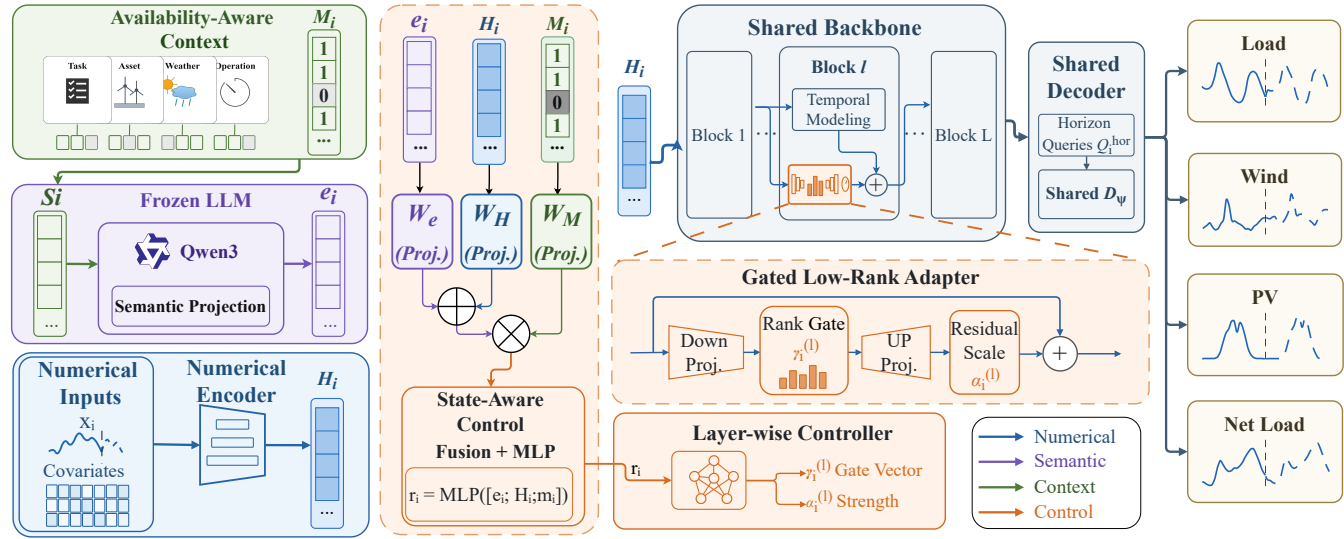

Availability-Aware Context
Task
Asset
Weather
Operation
M_i
1
1
0
1
S_i
Frozen LLM
Qwen3
Semantic Projection
e_i
Numerical Inputs
x_i
Covariates
Numerical Encoder
H_i
e_i
H_i
M_i
W_e (Proj.)
W_H (Proj.)
W_M (Proj.)
State-Aware Control Fusion + MLP
$r_i = MLP([e_i; H_i; m_i])$
r_i
Shared Backbone
H_i
Block 1
Block l
Temporal Modeling
Block L
Shared Decoder
Horizon Queries Q_i^{hor}
Shared D_ψ
Gated Low-Rank Adapter
Down Proj.
Rank Gate $\gamma_i^{(l)}$
UP Proj.
Residual Scale $\alpha_i^{(l)}$
Layer-wise Controller
$\gamma_i^{(l)}$ Gate Vector
$\alpha_i^{(l)}$ Strength
Load
Wind
PV
Net Load
Numerical
Semantic
Context
Control